

Mid Point Line Algorithm

Working Principle of Midpoint Line Algorithm

- The Midpoint Line Algorithm is used to draw a line on a raster display by deciding which pixel is closer to the actual line path.
- It works by checking the **midpoint** between two possible candidate pixels at each step.
- For a line with $0 < slope < 1$
- at each step, the next pixel can be either
 - $East(E) \rightarrow (x + 1, y)$
 - $North\ East(NE) \rightarrow (x + 1, y + 1)$

Midpoint Line Algorithm Formula

- Suppose the line endpoints are (x_1, y_1) and (x_2, y_2)
- Then $dx = x_2 - x_1$, $dy = y_2 - y_1$
- The initial decision parameter is

$$d_0 = dy - \frac{dx}{2}$$

- Decision rules
 - If $d < 0$ choose East pixel
 - $x_{k+1} = x_k + 1$
 - $y_{k+1} = y_k$
 - $d_{k+1} = d_k + dy$

If $d \geq 0$ choose North East pixel

- $x_{k+1} = x_k + 1$
- $y_{k+1} = y_k + 1$
- $d_{k+1} = d_k + dy - dx$

Algorithm MidpointLine(x_1, y_1, x_2, y_2)

```
{  
     $dx = x_2 - x_1$   
     $dy = y_2 - y_1$   
     $d = dy - dx/2$   
     $x = x_1$   
     $y = y_1$   
    putPixel( $x, y$ )  
    while ( $x < x_2$ )  
    {  
         $x = x + 1$   
        if ( $d < 0$ )  
             $d = d + dy$   
        else  
        {  
             $y = y + 1$   
             $d = d + dy - dx$   
        }  
        putPixel( $x, y$ )  
    }  
}
```

Algorithm MidpointLine(x_1, y_1, x_2, y_2)

1. Input the two endpoints (x_1, y_1) and (x_2, y_2)
2. Compute $dx = x_2 - x_1, dy = y_2 - y_1$
3. Set initial point $x = x_1, y = y_1$
4. Compute initial decision parameter $d = dy - \frac{dx}{2}$
5. Plot the starting pixel (x, y)
6. Repeat while $x < x_2$
 - Increment $x = x + 1$
 - If $d < 0$
 - Choose East Pixel
 - Update $d = d + dy$
 - Else
 - choose North-East pixel
 - Increment $y = y + 1$
 - Update $d = d + dy - dx$
 - Plot the new pixel (x, y)
7. Stop when $x = x_2$

Example Draw a line from (2,3) to (8,6)

- $dx = 8 - 2 = 6$ and $dy = 6 - 3 = 3$
- $d = dy - \frac{dx}{2} = 3 - \frac{6}{2} = 0$

Step	Current Point	Choice	Next Point	New d
0	(2, 3)	Start	(2, 3)	0
1	(2, 3)	NE	(3, 4)	$(0 + 3 - 6 = -3)$
2	(3, 4)	E	(4, 4)	$(-3 + 3 = 0)$
3	(4, 4)	NE	(5, 5)	$(0 + 3 - 6 = -3)$
4	(5, 5)	E	(6, 5)	$(-3 + 3 = 0)$
5	(6, 5)	NE	(7, 6)	$(0 + 3 - 6 = -3)$
6	(7, 6)	E	(8, 6)	$(-3 + 3 = 0)$